

EF5342 Financial Systems, Markets and Instruments

Week 3 — Bond Markets and Bond Pricing

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Bond Markets and Bond Pricing

- Bond characteristics
- Treasury bonds, Munis, and corporate bonds
- Bond pricing and yield to maturity
- Interest rate risk and duration

Fixed-Income Securities

Fixed-income security (a.k.a. debt security, bond) promises a fixed stream of income, or a stream determined by a pre-specified formula

Key issues:

- How to value future payments (time value of money)
- Risk that the promise will not be kept (credit risk)

Two main payment streams:

Zero-coupon bonds A single payment at maturity — the face (par) value

Coupon bonds Pay $\$C$ each period (typically every six months) up to maturity, when they pay $\$C + \text{face value}$

Fixed-Income Payoffs

Zero-coupon bond: nothing until $t + m$, then the face value (e.g., \$1,000).

Coupon bond: $\$C$ at $t + 1$, $\$C$ at $t + 2$, ..., and $\$C + \$1,000$ at maturity $t + m$.

Coupon rate = % of par value paid every coupon period, usually quoted annually; coupons can be paid more frequently.

- Example: an 8% semi-annual coupon bond pays \$40 every 6 months

	t	$t+1$	$t+2$	...	$t+m$
Zero					1,000
Coupon		C	C	...	$1,000 + C$

Coupon Bonds

Before electronic registration, bonds were issued as physical certificates with detachable coupons — every six months the holder tore off one coupon in exchange for the interest payment.

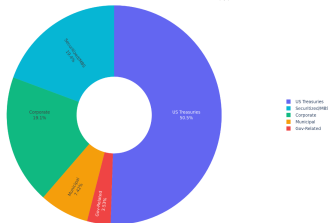


Major Types of Bonds

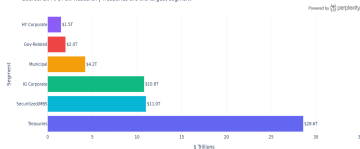
Bonds are long-term debt obligations issued by corporations and governments (or government agencies):

- Treasury notes (T-notes) and bonds (T-bonds)
- Municipal bonds (Munis)
- Corporate bonds (investment-grade and high-yield)
- Mortgage-backed securities (MBS) and asset-backed securities (ABS)

US Fixed-Income Market Composition (~\$58T, 2025)
Source: SIFMA / Pew Research | Treasuries dominate at ~49%



US Fixed-Income Market by Segment (~\$58T, 2025)
Source: SIFMA, Pew Research | Treasuries are the largest segment



Treasury Notes and Bonds

Issued by the U.S. Treasury to finance the national debt and other government expenditures.

- **Default-risk free:** backed by “full faith and credit” of the U.S. federal government
- **Low returns:** low interest rates reflect minimum default risk
- **Interest rate risk:** longer maturity $\Rightarrow$ wider price fluctuations than T-bills when rates change
- **Liquidity risk:** very active secondary markets, but older (“off-the-run”) issues are less liquid than newly issued (“on-the-run”) ones
- **Inflation risk:** inflation may erode the purchasing power of future nominal payments

T-notes: original maturities of 2, 3, 5, 7, and 10 years; T-bonds: > 10 years. Prices are quoted as a percentage of par (usually \$1,000).

Interest Rate Risk: Treasury Bonds vs. Notes

The total-return index of Treasury **bonds** with 20+ years of maturity (blue) is much more volatile than Treasury **notes** with 3–7 years of maturity (orange).

HOME > IDCOT20TR - INDEX

ICE U.S. Treasury 20+ Year TR Index

103.93 ↓ 2.01% -2.13 5Y

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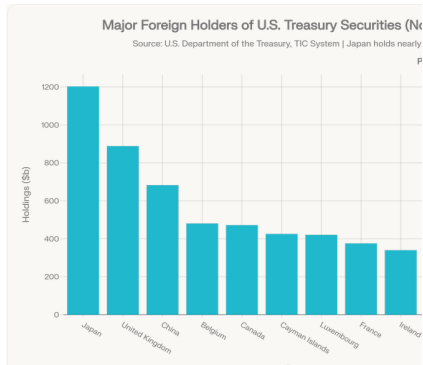
1D 5D 1M 6M YTD 1Y 5Y MAX



Source: Google Finance

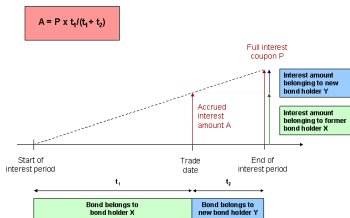
Major Foreign Holders of U.S. Treasury Securities

As of June 2026, Japan is the largest foreign holder of U.S. Treasury securities (\$1,116 billion), followed by the United Kingdom (\$940 billion) and China (\$633 billion — its lowest level since 2008); total foreign holdings are \$9.29 trillion (Treasury TIC). The U.K. has overtaken China as the second-largest holder.



Accrued Interest

Accrued interest is the portion of the coupon payment accrued between the last coupon payment and the settlement date. It must be paid by the *buyer* of a bond to the *seller* if the bond is purchased between coupon payment dates.



- Accrued interest on T-bonds is calculated as:

- $$\text{Accrued interest} = \frac{C}{2} * \frac{\text{Actual \# of days since last coupon}}{\text{Actual \# of days in coupon period}}$$

- The price of the T-bond with accrued interest is called the **invoice price**, while the price without accounting for accrued interest is the **flat price**
- In US, bond price quoted in newspaper is “flat price”, but bond trade settled at “invoice price”
- **Invoice price = Flat price + Accrued Interest**

Accrued Interest: Example

You buy a 6% coupon \$1,000-par T-bond 59 days after the last coupon payment; settlement occurs in two days, so you become the owner **61 days** after the last coupon, with **121 days** remaining until the next coupon. The bond's flat price is quoted at 120.593. What is the invoice price?

- Flat price: $120.593\% \times \$1,000 = \$1,205.93$
- Accrued interest: $\frac{61}{61+121} \times \$30 = \$10.05$
- Invoice price: $\$1,205.93 + \$10.05 = \mathbf{\$1,215.98}$

$$\text{Accrued Interest} = \frac{\$60}{2} \times \frac{61}{(121 + 61)} = \$10.05$$

Treasury Inflation-Protected Securities (TIPS)

TIPS protect investors against inflation by adjusting the bond's **principal daily** based on changes in the CPI-U. The coupon rate is fixed — but applied to the inflation-adjusted principal, so coupons rise with inflation.

- Example: \$1,000 par, 3% coupon, 10% inflation $\Rightarrow$ principal adjusts to \$1,100; coupon becomes $3\% \times \$1,100 = \33 rather than \$30
- TIPS provide a **real** yield; nominal Treasuries a nominal yield
- At maturity, investors receive the *greater* of adjusted principal or original face — deflation protection

Decomposition: Nominal Treasury yield = TIPS real yield + breakeven inflation rate.
Expected inflation turned negative in the 2008 GFC period.



How the Treasury Market Works

The U.S. Treasury conducts regular auctions through TAAPS. Three types of bidders:

Primary dealers 24 designated counterparties (e.g., JPMorgan, Goldman Sachs, Citi), required to bid at all auctions at reasonably competitive prices — market-makers of last resort

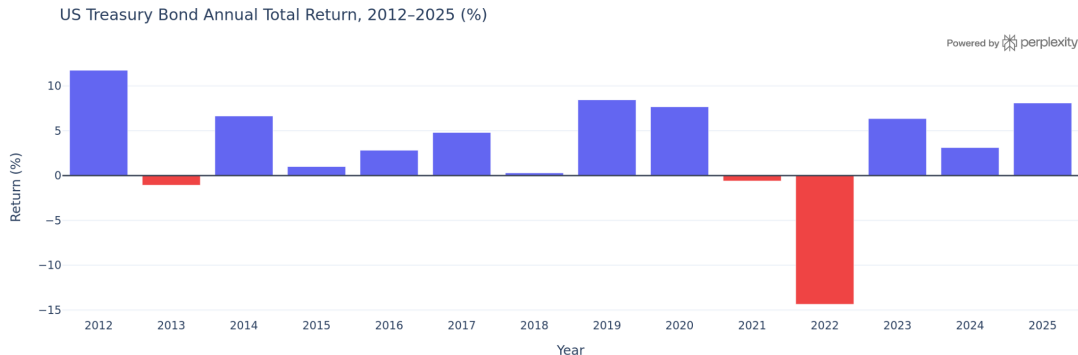
Direct bidders Institutional investors (pension funds, central banks) accessing auctions directly through TAAPS

Indirect bidders Foreign official institutions and investors routing through intermediaries — often read as a proxy for foreign central bank demand

The secondary market is an OTC dealer market with primary dealers quoting bid–ask spreads; average daily trading volume was roughly \$1.0 trillion in 2025 (SIFMA, up 16% year-over-year) — the world's most liquid market.

Historical Returns of U.S. Treasury Bonds

2022 was catastrophic: -14.36% total return for U.S. Treasuries — the worst single year in modern U.S. bond market history. The Bloomberg U.S. Aggregate fell -13.01% . This reflected the fastest tightening cycle since the 1980s. Returns were positive again in 2025 amid historically tight credit spreads.

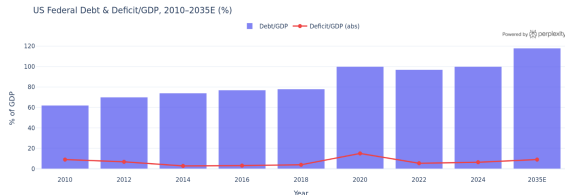


Fiscal Dynamics and the Bond Market

The U.S. federal deficit reached **\$1.78 trillion in fiscal year 2025 (5.9% of GDP)**; debt held by the public stood near 99.8% of GDP, and net interest exceeded \$1 trillion for the first time (CBO). These dynamics affect yields through two channels:

Supply effect Persistent large issuance requires the market to absorb more Treasury supply; if demand does not scale proportionally, yields must rise to attract buyers

Term-premium channel Fiscal uncertainty raises the risk premium investors demand for holding long-duration government bonds — lifting long-end yields independently of policy-rate expectations



Municipal Bonds

Debt securities issued by state and local governments to finance long-term projects (roads, schools, airports. . .). Attractive because interest income on most Munis is **exempt from federal income tax** (and often from state and local taxes for in-state residents).

General obligation (GO) bonds Backed by the full faith and credit of the issuing municipality

Revenue bonds Sold to finance specific revenue-generating projects

Question: which type is riskier?

Municipal Bond Yields

Because Munis are tax-exempt, compare after-tax yields. For a corporate bond:

$$i_a = i_b(1 - t) \quad \text{where } i_a = \text{after-tax yield, } i_b = \text{before-tax yield, } t = \text{marginal tax rate}$$

Equivalently, convert the Muni yield to an equivalent before-tax yield: $i_b = i_a/(1 - t)$.

Examples (28% tax bracket):

- After-tax yield of a 6% corporate yield: $i_a = 6\% \times (1 - 0.28) = \mathbf{4.32\%}$
- Corporate yield equivalent to a 4.5% Muni: $i_b = 4.5\% / (1 - 0.28) = \mathbf{6.25\%}$

Corporate Bonds

Long-term debt obligations issued by corporations; most trade over-the-counter in a network of bond dealers. Listings include issuer name, coupon rate, maturity date, price, yield, and credit ratings.

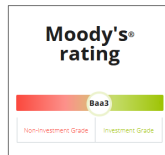
BOEING CO

Symbol: BA4983330 CUSIP: 097023CW3 Bond Type: CORP

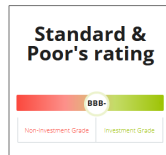
[TRADE HISTORY](#)[ADD TO WATCHLIST](#)

Coupon Rate	5.805% ▼
Maturity Date	5/1/2050 ▼
Last Trade Price	\$100.07 ▼
Last Trade Yield	5.79% ▼
Last Trade Date	1/15/2026 ▼
Callable	Y ▼
Next Call Date	11/1/2049 ▼
Price/Yield Chart	^

Credit Rating



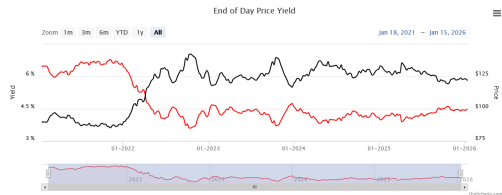
Last rated on 2025-12-11



Last rated on 2020-04-30

Bond Prices and Yields: An Inverse Relationship

Price/Yield Chart



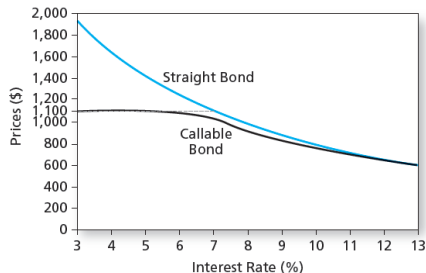
Convertible bonds give holders an option to exchange each bond for a specified number of common shares of the issuer.

- Example: a \$1,000-par convertible exchangeable into 40 shares (stock at \$20): converting now yields $40 \times \$20 = \$800 < \text{par}$ — not profitable now; profitable once the stock rises above \$25
- The option is valuable $\Rightarrow$ convertibles offer *lower* yields than comparable non-convertibles

Callable Bonds

A **call provision** lets the issuer repurchase the bond at a pre-specified (call) price after a certain time.

- Bonds are usually called when interest rates *drop*, so the issuer can refinance at lower rates
- Bondholders dislike calls — they must reinvest at lower rates $\Rightarrow$ callable bonds offer *higher* yields than comparable non-callables
- The callable bond's value is **capped by the call price** as interest rates decrease



Corporate bonds without options attached are called **straight bonds**.

Credit Risk and Bond Ratings

Are coupons/par guaranteed? **No** — default is possible. Recent defaults: Enron, WorldCom, Lehman; sovereigns Russia (1998), Argentina (2001), Greece (2012); USD bonds of China property developers like Evergrande (2021). Three major rating agencies: **Moody's, S&P, Fitch**. Bonds are rated by perceived default risk and categorized as **investment grade** or **speculative grade** (junk / high-yield).

- Speculative grade $\Rightarrow$ higher default probability, compensated by higher yields
- **Fallen angel**: downgraded from IG to HY; **rising star**: upgraded HY to IG (expanded buyer base)

	MOODY'S	STANDARD & POORS
INVESTMENT GRADE		
Highest quality (Best quality, smallest degree of investment risk)	Aaa	AAA
High Quality (Often called high-grade bonds)	Aa	AA
Upper medium grade (Many favourable investment attributes)	A	A
Medium grade (Neither highly protected nor poorly secured)	Baa	BBB
SPECULATIVE GRADE		
Somewhat speculative (Some speculative elements)	Ba	BB
Speculative (Generally lack characteristics of a desirable investment)	B	B
Highly speculative (Bonds of poor standing)	Ca	CC
Most speculative (Poor prospects)	C	C
Imminent default (Extremely poor prospects)	C	C
	C	D

Bond Pricing

A bond's value is the sum of the present values of its future payment streams. Three components: future cash flows, the interest rate (discount rate), and price (present value).

Zero-coupon bond:

$$P = \frac{FV}{(1+r)^T}$$

Coupon bond (pays \$C each period for T periods):

$$P = \sum_{t=1}^T \frac{C}{(1+r)^t} + \frac{FV}{(1+r)^T}$$

where r is the discount rate investors require to hold the bond. Coupon rates and r are quoted *annually* (APR convention); most bonds pay coupons semi-annually.

Semi-Annual Compounding

With semi-annual coupons, adjust both the payment and the discount rate. One dollar grows to:

$$\$(1 + \frac{r}{2}) \text{ after 6 months,} \quad \$(1 + \frac{r}{2})^2 \text{ after 1 year,} \quad \$(1 + \frac{r}{2})^{2T} \text{ after } T \text{ years}$$

So the semi-annual bond pricing formula (coupon halved to $C/2$, discount rate $r/2$, $2T$ payments):

$$P = \sum_{t=1}^{2T} \frac{C/2}{(1 + \frac{r}{2})^t} + \frac{FV}{(1 + \frac{r}{2})^{2T}}$$

Bond Pricing: Example

Price a bond that pays a **10% coupon** and matures in **12 years**; the interest rate is **8%**.
(Unstated face value is \$1,000 by convention.)

- Too many coupons to add up one by one — use the annuity formula:

$$\sum_{t=1}^T \frac{C}{(1+r)^t} = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

- Semi-annual coupon bond: $r \rightarrow 4\%$, $T \rightarrow 24$ periods, $C \rightarrow \$50$:

$$P = \frac{50}{0.04} \left(1 - \frac{1}{1.04^{24}} \right) + \frac{1000}{(1+0.04)^{24}} = \$1,152.47$$

Excel: =PV(0.04, 24, 50, 1000) **Financial calculator:** $I = 4$, $PMT = 50$, $FV = 1000$,
 $T = 24 \Rightarrow PV = -1152.47$

Interest Rates and Bond Prices

Same 10%-coupon, 12-year bond at different required rates:

$$r = 12\% : P = \frac{50}{0.06} \left(1 - \frac{1}{1.06^{24}} \right) + \frac{1000}{1.06^{24}} = \$874.50 \quad r = 10\% : P = \$1,000 \text{ (par)}$$

Interest rate	Bond price	
12%	874.50	discount
10%	1,000	par
8%	1,152.47	premium

VERY important:

- **Inverse relationship** between bond price and interest rate (if r falls, P rises)
- The price–rate relationship is **non-linear** (convex)
- **Premium bond:** $C > r \Rightarrow V_b > \text{Par}$; **discount bond:** $C < r \Rightarrow V_b < \text{Par}$; **par bond:** $C = r \Rightarrow V_b = \text{Par}$

Yield to Maturity

Yield to maturity (YTM) The interest rate y at which the bond's present value equals its price — the bond's IRR

Roughly, the average annual return from buying at the market price and holding to maturity.
For semi-annual bonds, solve:

$$P = \sum_{t=1}^{2T} \frac{C/2}{(1 + \frac{y}{2})^t} + \frac{FV}{(1 + \frac{y}{2})^{2T}}$$

Calculable in Excel or with a financial calculator.

Yield to Maturity: Example

Consider a semi-annual bond: $T = 2$ years, $C = \$20$ per period, $P = 909.2$, $FV = \$1,000$.
Solve:

$$909.2 = \sum_{t=1}^4 \frac{20}{(1 + \frac{y}{2})^t} + \frac{1000}{(1 + \frac{y}{2})^4} \implies y = \mathbf{6.94\%} \text{ (annualized)}$$

What Factors Affect Bond Yields?

Default risk The most important factor: higher default risk $\Rightarrow$ investors require higher yields

Maturity Longer-maturity bonds are more sensitive to rate changes, so typically have higher yields

Liquidity Less liquid bonds pay higher yields to compensate for trading costs

Convertibility Convertibles have *lower* yields — investors get the valuable conversion option

Callability Calls are typically exercised when rates fall $\Rightarrow$ investors demand higher yields on callable bonds

Credit Spreads

Corporations have larger default risk and higher yields than Treasuries of similar maturity — creating a **credit spread**.

- Historically, IG spreads average ~130 basis points; HY spreads ~450 bps
- COVID shock (March 2020): IG spreads jumped to ~372 bps, HY to over 1,000 bps before central-bank intervention compressed them; as of end-August 2026 spreads remain historically tight (HY OAS 2.63% vs. a long-run average near 475 bps)

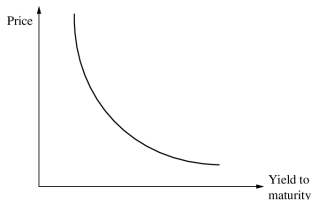


Interest Rate Risk

Bond prices are sensitive to interest rates:

- **Inverse relationship:** rates rise $\Rightarrow$ prices fall (rising rates are bad news for bond investors)
- The relationship is **convex**, not linear — but for small changes a linear approximation works
- **Long-term** bonds are more sensitive than short-term (the coupon is locked in for longer)
- **Higher-coupon** bonds pay more cash flow early, so they are less sensitive

We want a measure of “how long-term” a bond is that captures both maturity and coupon effects.



Duration measures the average maturity of a bond's promised cash flows: the weighted average of times to each coupon or principal payment, where the weight on each payment time is the proportion of bond value it accounts for.

$$D = \sum_{t=1}^T t \cdot w_t, \quad w_t = \frac{CF_t / (1+y)^t}{P}, \quad \sum_t w_t = 1$$

where P = bond price, CF_t = cash flow at end of period t , y = yield to maturity.

Duration: Examples 1 and 2

- 6% Coupon, 4-year maturity, annual payment bond with a 5% ytm
- $D = \sum_{t=1}^T t * w_t$
- $w_t = \frac{CF_t / (1+y)^t}{P}$

- 3% Coupon, 4-year maturity, annual payment bond with a 5% ytm
- $D = \sum_{t=1}^T t * w_t$
- $w_t = \frac{CF_t / (1+y)^t}{P}$

Worked calculation: Part 1 finds the present value of each cash flow; Part 2 takes the weighted average of payment times. Slide results: **Example 1:** $D = 3.6793$ years; **Example 2** (lower coupon): $D = 3.8213$ years.

Duration: Example 3

- 6% Coupon, 6-year maturity, annual payment bond with a 5% ytm
- $D = \sum_{t=1}^T t * w_t$
- $w_t = \frac{CF_t / (1+y)^t}{P}$

Duration = 5.2342 years.

With a **longer maturity**, duration is longer and price volatility larger. Of the two effects, **maturity is the predominant** determinant of duration.

Duration of Zero-Coupon Bonds

A zero-coupon bond makes a single payment, so its duration equals its maturity:

$$D = 1 \times \frac{CF_1/(1+y)}{P} = T \quad (\text{one payment at } T)$$

E.g., a one-year zero: $D = 1 \cdot \frac{1000/1.10}{1000/1.10} = 1$.

- Coupon bonds always have $D < T$
- The higher the coupon rate, the more weight on early payments and the shorter the duration

What Determines Bond Duration?

- Coupon rate** Higher coupon $\Rightarrow$ *shorter* duration: cash flows arrive earlier and carry higher weights in the calculation
- Maturity** Longer maturity $\Rightarrow$ *longer* duration: generalizes the idea that long-term bonds are more rate-sensitive than short-term bonds
- Yield to maturity** Higher YTM $\Rightarrow$ *shorter* duration: later cash flows are discounted more heavily, shrinking their weights

Duration and Interest Rate Risk

Duration gives a linear approximation to the percentage price change:

$$\frac{\Delta P}{P} \approx -D \times \frac{\Delta y}{1+y}$$

Example. A 2-year bond with 12% coupon at a 10% yield has $D = 1.8859$ years. If rates rise from 10% to 12% ($\Delta y = 0.02$):

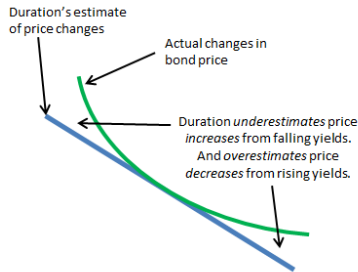
$$\frac{\Delta P}{P} \approx -1.8859 \times \frac{0.02}{1.10} = -3.43\%$$

Duration-Based Prediction Errors

Duration predicts price changes accurately only for **small** rate changes; day to day it works quite well. But when rates move significantly — e.g., Fed funds-rate adjustments — predicted pricing errors become significant.

- Consider the four-year, \$1,000 face value bond with a 10 percent coupon paid semiannually and an 8 percent YTM. This bond has a duration of 3.42 years, and its current price is \$1,067.34. If YTM rises from 8 percent to 10 percent, the duration model predicts that the bond price will fall by 6.577 percent; that is:
$$\frac{\Delta P}{P} = -D * \frac{\Delta y}{1 + \frac{y}{2}} = -3.42 * \left(\frac{2\%}{1.04}\right) = -6.577\%$$
or from a price of \$1,067.34 to \$997.14.
- However, using bond pricing formula to calculate the bond's true price after a rise in yield to 10 percent, we find the true price is \$1,000.
- The actual fall in price is less than the duration predicted fall by \$2.86.
- The prediction errors arise because bond price changes are not linear with respect to interest rate change.

Bond
Price



Yield

The price–rate relationship is non-linear; **convexity (CX)** measures the change in the slope of the price–yield curve around rate level R , adding curvature to the duration estimate:

$$\frac{\Delta P}{P} \approx -D \times \frac{\Delta y}{1+y} + \frac{1}{2} CX \times (\Delta y)^2$$

- The convexity term means the bond price is always *above* the duration-model prediction
- Convexity is desirable: all else equal, more-convex bonds outperform less-convex bonds when yields move significantly in *either* direction

Key Takeaways

Bond basics Fixed-income securities promise a stream of cash flows. Key types: Treasury (default-free), municipal (tax-exempt), corporate (default risk)

Bond features TIPS: principal adjusts with inflation, providing a real yield. Convertible: exchangeable for stock, lower yields. Callable: issuer redeems early, higher yields

Bond pricing and yields $\text{Price} = \text{PV of coupons} + \text{PV of par (semi-annual discounting)}$. Bond prices are **inversely** related to interest rates

Interest rate risk Long-term and low-coupon bonds are most rate-sensitive. Duration measures average maturity and price sensitivity